CHAPTER – 01

INTRODUCTION

1.1 – Project Overview

Heart diseases continue being one of the main causes of fatalities all around the world, therefore, making timely detection and evaluation of risks related to this condition crucial. The purpose of this project is to use Tableau to visualize patients' health data associated with heart disease, focusing on several factors, including age, gender, cholesterol level, blood pressure, chest pain type, and maximum heart rate.

By creating the dashboard and using different visual aids in terms of charts, bar diagrams, and KPIs, the audience will have the opportunity to discover some hidden patterns, which can help with finding out how certain factors contribute to heart diseases and what health trends should be considered by specialists.

Project Objectives:

- To analyze the prevalence of heart diseases among the target population.
- To establish vulnerable demographic segments based on certain criteria.
- To find out whether cholesterol and blood pressure affect heart diseases negatively.
- To examine the connection between chest pain types and exercise-induced angina and diagnoses.
- To offer analytical insights into health data based on the developed visualizations.
- Dashboard Highlights

The Tableau dashboards represent various visual elements and KPIs that are used to identify patterns and get insights into how often heart diseases occur within certain groups of patients.

Key Metrics:

- Total Number of Patients
- Cases of Heart Diseases
- Percentage of Patients with Heart Diseases
- Average Age
- Average Cholesterol Level
- Average Resting Blood Pressure
- Insights
- Heart diseases are likely to be more common at older ages.
- Heart diseases are more prevalent among male patients compared to females.
- Higher cholesterol and blood pressure levels are typical for affected patients.

- There is a certain correlation between heart diseases and certain chest pain types.
- Exercise-induced angina is likely to be a predictor of heart diseases.